

Statistics Hybrid Practice Set – S16

Main Focus

This set targets the weak area from S15: **interpretation and choosing the correct measure**, not just calculating formulas.

Topics covered

- Mean vs Median
- Range vs IQR
- Standard Deviation
- Coefficient of Variation
- Quartiles
- Five Number Summary
- Outliers
- Box Plot interpretation
- Skewness
- Choosing the right statistical measure

Q1. Mean vs Median

A company's daily sales are:

₹900, ₹950, ₹1,000, ₹1,050, ₹1,100, ₹1,150, ₹5,000

Answer:

1. Calculate the mean.
2. Calculate the median.
3. Which measure better represents typical daily sales?
4. Explain specifically **how ₹5,000 affects the mean.**

Q2. Range vs IQR

Dataset:

5, 7, 8, 9, 10, 11, 12, 13, 50

Find:

1. Range
2. Q1
3. Q3
4. IQR
5. Which measure, **Range or IQR**, is less affected by the extreme value 50?
6. Why?

Q3. Standard Deviation Interpretation

Two datasets have the same mean:

Dataset	Mean	Standard Deviation
A	100	5
B	100	20

Answer:

1. Which dataset has greater spread?
 2. Which dataset is more consistent?
 3. Explain what the difference between $SD = 5$ and $SD = 20$ tells you.
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Q4. Coefficient of Variation

Three companies have:

Company	Mean Sales	Standard Deviation
A	₹40,000	₹4,000
B	₹60,000	₹9,000
C	₹100,000	₹5,000

Calculate the CV for all three.

Then answer:

1. Which company has the **lowest relative variability**?
 2. Rank the companies from **most consistent to least consistent**.
 3. Why would comparing standard deviation alone give the wrong impression here?
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Q5. Five Number Summary

Dataset:

4, 6, 7, 9, 10, 12, 15, 18, 20

Find:

1. Minimum
2. Q1
3. Median
4. Q3
5. Maximum
6. IQR
7. Range

Q6. Outlier Detection

Using the dataset from Q5:

4, 6, 7, 9, 10, 12, 15, 18, 20

Calculate:

1. Lower Fence
 2. Upper Fence
 3. Are there any outliers?
 4. If a new value **40** is added, would it be an outlier? Show why.
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Q7. Box Plot Interpretation

A dataset has this Five Number Summary:

Statistic	Value
Minimum	10
Q1	20
Median	25
Q3	45
Maximum	90

Answer:

1. What is the IQR?
2. What percentage of observations lie between 20 and 45?
3. Which half of the box has greater spread?
4. Is the median closer to Q1 or Q3?
5. What does this suggest about the distribution?
6. What additional feature of the box plot supports your conclusion?

Q8. Identify the Incorrect Conclusion

A student says:

“Dataset A has a lower standard deviation than Dataset B, so Dataset A must also have a lower coefficient of variation.”

Is this always true?

Answer:

1. Yes or No?
2. Explain why.
3. Give a simple example using different means.

Q9. Choose the Best Method

For each scenario, choose the **most appropriate measure or visualization**.

A.

You want to represent the typical income when a few people have extremely high incomes.

B.

You want to compare relative consistency between two departments with very different average salaries.

C.

You want to detect possible outliers and compare distributions of three departments visually.

D.

You want to measure the spread of the middle 50% of observations.

Answer with the correct measure/visualization and a short reason.

Q10. Mixed Data Analyst Case

A company compares delivery performance:

Company	Mean Time	SD	Median
A	30 min	6 min	29 min
B	50 min	5 min	50 min

Answer:

1. Calculate CV for A and B.
2. Which company has lower **absolute variability**?
3. Which company has lower **relative variability**?
4. If consistency relative to average delivery time matters, which company performs better?
5. Why are the answers to Q2 and Q3 different?
6. Based on the mean and median, which company shows more evidence of skewness?